**Project description:**

The goal of the project is to apply the Big Data Analytic project cycle (that you studied in lecture) on your selected dataset you like. In the project lifecycle phases, you will use and apply the methods and techniques studied in both lecture and lab. The total marks of the course project will be evaluated according to: (i) project deliverables and (ii) individual oral discussion.

<u>The project steps should meet the following:</u>

1. Search for a real dataset with multiple attributes (a minimum of 10 relevant attributes)
2. Understand the dataset and scientifically define your project objectives w.r.t. data science objectives.
3. Apply data preprocessing/cleaning methods (if needed)
4. Use Hypothesis Testing
5. Apply the EDA using different data visualization charts. Select the appropriate chart that allow you to observe more knowledge about your dataset.
6. Mention your comments and observations on your visualizations.
7. Apply any data analytics technique you studied in the course (e.g., ID3, Regression, Apriori, Kmeans, etc.)

**Deliverables:**

1. Dataset (.CSV file)
2. Code files (R script)
3. Project Documentation that includes:
   a. The project description
   b. The dataset and variables description
   c. The problem definition and project objectives
   d. The data visualization graphics with observations and interpretations of each chart
   e. Any applied data cleaning or transformation methods (if applied)
   f. Any applied Hypothesis and the test results and interpretations
   f. The dataset preparation in terms of machine learning (training set, learning set)
   g. The used data analytics techniques (e.g. ID3, Regression, Apriori, Kmeans, etc.), with justification for your choice
   h. The performance measures used and the evaluation of analytical technique
   i. Discussion/Quantification for relevant project findings for your project.

4. Each team is required to **design and present a visual project poster** that effectively summarizes the **Big Data Analytics Lifecycle** applied in your project — from **data collection** to **deployment**.

   **Poster Specifications**

   - **Format:** A3 size

- **Orientation:** Portrait
- **Submission:** Printed hard copy
- **Purpose:** To visually communicate the key stages, methodologies, and outcomes of your Big Data project in a concise and professional manner.

**Poster Content Requirements**

**The poster must include the following sections:

1. **Project Title and Team Information**
2. **Overview of Dataset and Problem Statement and objectives**
3. **Key Visualizations and their analysis**
4. **Analytical methods used and their analysis**
5. **Summary of Results and Insights**